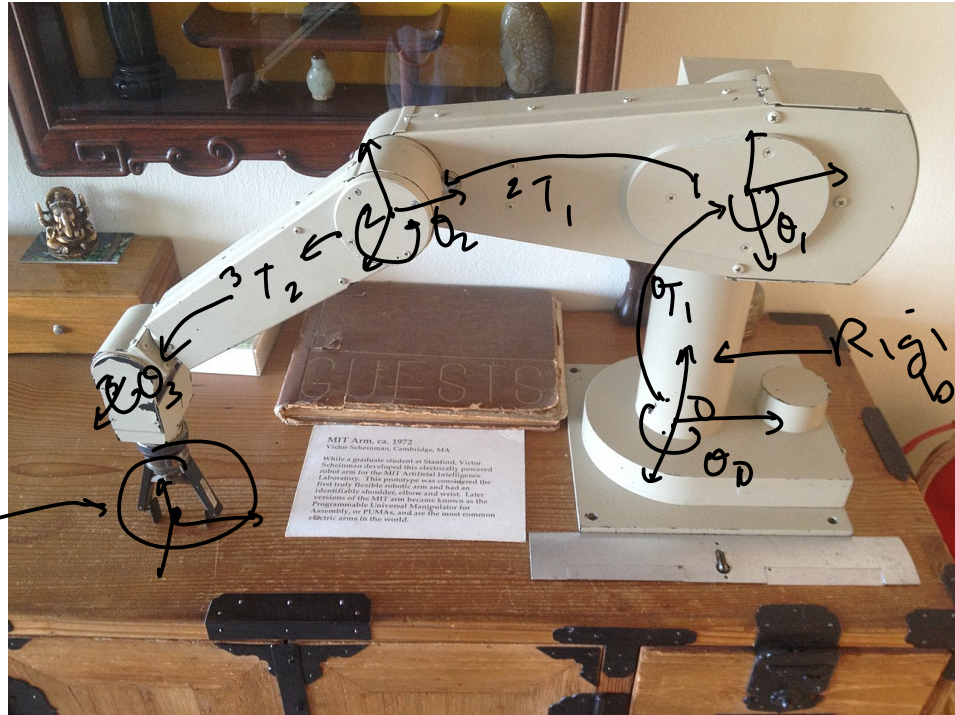


Forward and inverse kinematics



end effector

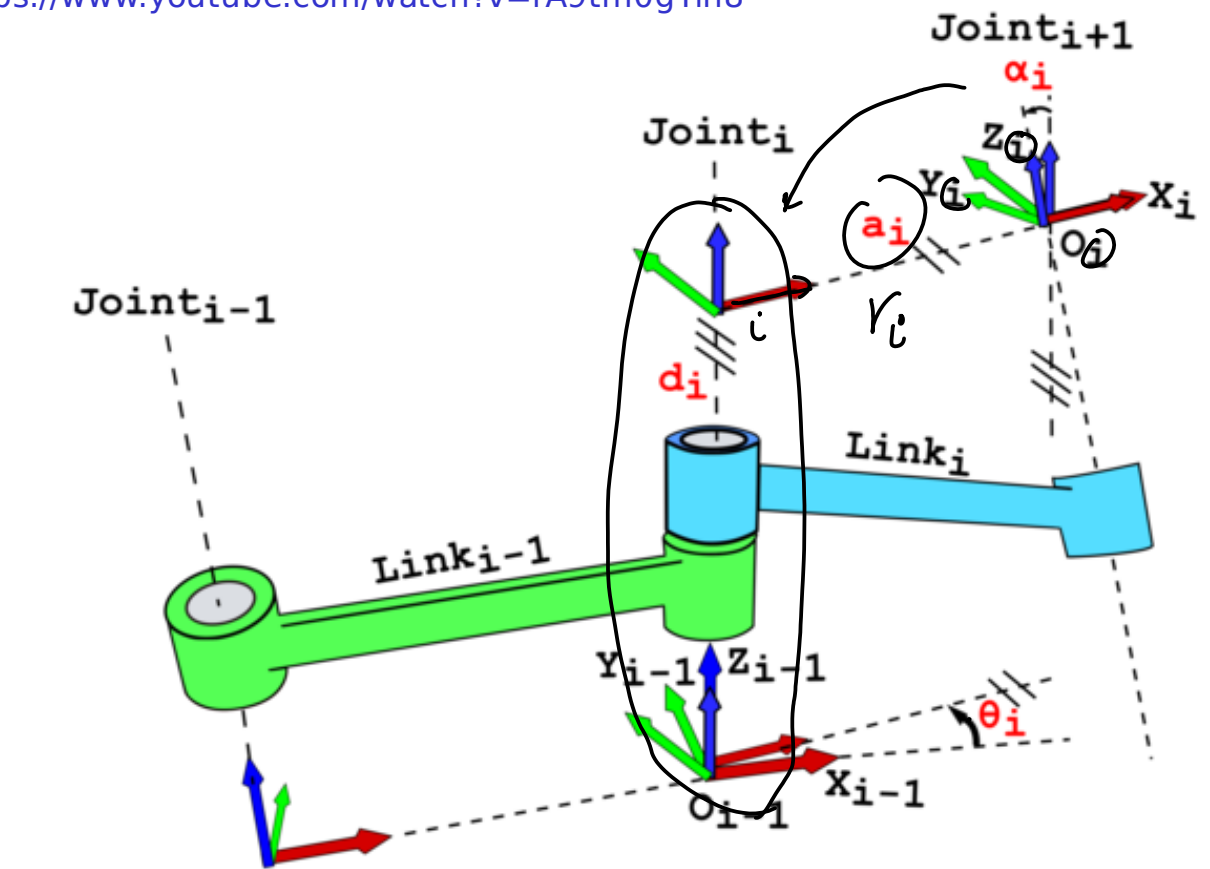
$${}^4T_0 = {}^4T_3(\theta_3) {}^3T_2(\theta_2) {}^2T_1(\theta_1) {}^1T_0(\theta_0)$$

Pose of end effector $\xleftarrow{\text{Fwd Kinematics}}$ Joint angles

Pose of end effector $\xrightarrow{\text{Inverse Kinematics}}$

Denavit Hartenberg parameters

<https://www.youtube.com/watch?v=rA9tm0gTln8>



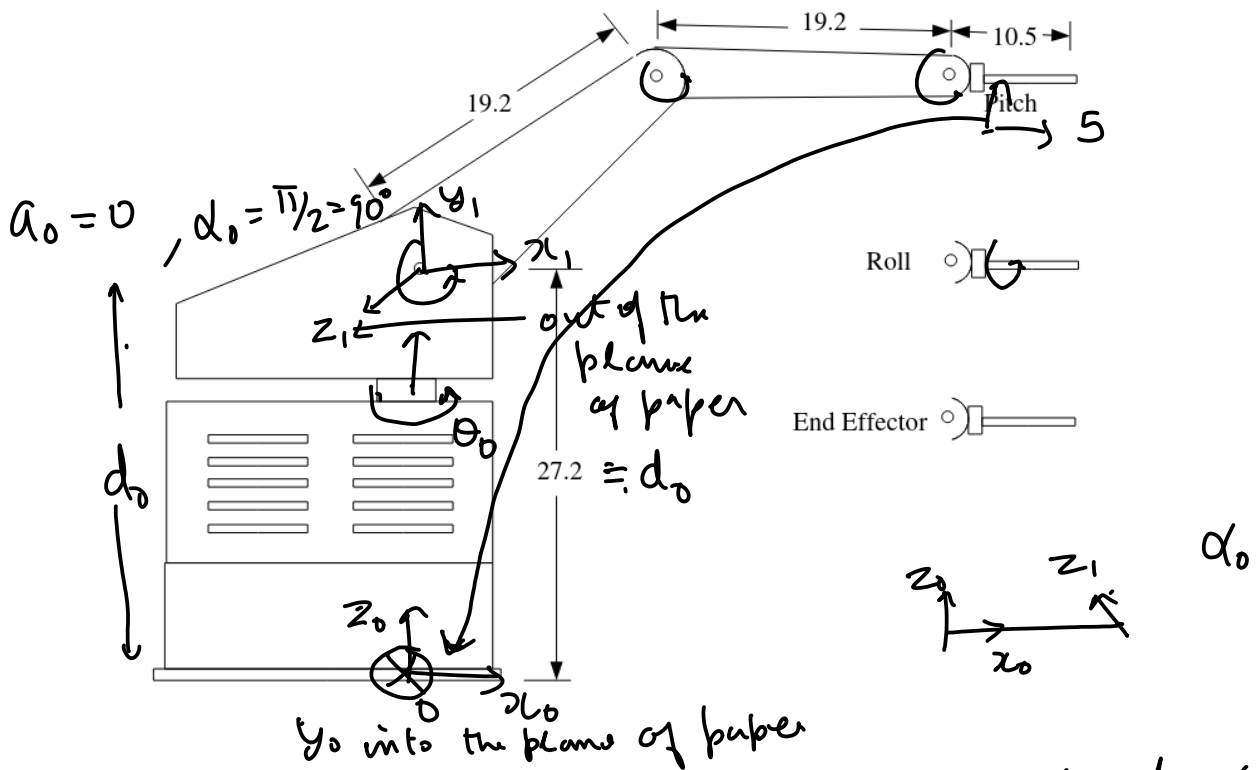
$${}^{i-1}T_i = {}^{i-1}T_{z_i} {}^{z_i}T_{x_i}$$

$$= \begin{bmatrix} R_z(\theta_i) & \underline{d_i} \\ 0 & 1 \end{bmatrix} \begin{bmatrix} R_x(\alpha_i) & \underline{a_i} \\ 0 & 1 \end{bmatrix}$$

$${}^{i-1}T_i = \begin{bmatrix} \cos \theta_i & -\sin \theta_i & 0 \\ \sin \theta_i & \cos \theta_i & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} \cos \theta_i & -\sin \theta_i & 0 & 0 \\ \sin \theta_i & \cos \theta_i & 0 & 0 \\ 0 & 0 & 1 & d_i \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & a_i \\ 0 & \cos \alpha_i & -\sin \alpha_i & 0 \\ 0 & \sin \alpha_i & \cos \alpha_i & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

${}^{i-1}T_i$ ← D-H params $(\theta_i, d_i, \alpha_i, a_i)$

Find the Denavit Hartenberg Parameters of the Lab volt arm

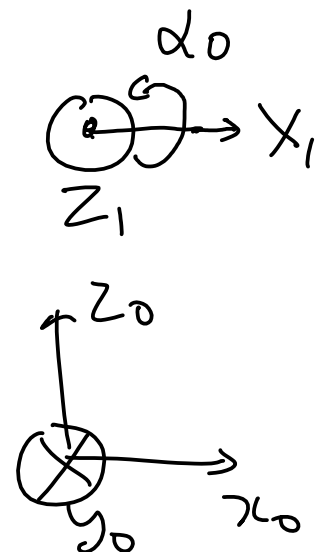


Pose = Position + orientation of the end effector (5)

$${}^0T_5 = {}^0T_1 {}^1T_2 {}^2T_3 {}^3T_4 {}^4T_5$$

Find DH params of the arm

i	d_i	θ_i	r_i/a_i	α_i
0	27.2	var	0	$\pi/2 = 90^\circ$
1	0	var	19.2	0
2	0	var	19.2	0
3				



α_0 is the rotation around x_0 so that

Z_0 aligns with Z_1